

PAULS VALLEY, OK, USA

WMO: 722173

Lat: 34.711N

Lon: 97.223W

Elev: 295

StdP: 97.83

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53951

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.9	-6.2	-15.9	1.0	-6.7	-12.5	1.3	-2.2	11.7	10.7	10.7	9.7	3.0	340	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	37.9	23.9	37.0	23.9	35.1	24.2	26.2	33.2	25.7	32.8	25.2	32.4	4.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.8	28.7	23.8	19.3	28.5	23.0	18.4	28.0	82.4	33.3	80.4	32.9	78.5	32.0	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.6	9.1	8.2	-12.3	39.8	2.3	2.0	-14.0	41.2	-15.3	42.4	-16.6	43.5	-18.2	44.9
			-12.8	27.2	2.1	0.8	-14.3	27.8	-15.5	28.2	-16.7	28.7	-18.3	29.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	5.3	7.4	12.8	16.8	21.1	26.6	28.5	28.2	24.3	17.5	11.6	6.0	(6)
(7)		DBStd	9.50	5.84	6.46	5.48	4.74	4.11	2.67	2.72	3.21	3.61	5.06	5.28	5.59	(7)
(8)		HDD10.0	519	168	114	33	7	1	0	0	0	6	42	148		(8)
(9)		HDD18.3	1669	406	308	183	82	21	0	0	3	77	208	383		(9)
(10)		CDD10.0	3159	21	43	121	211	346	499	573	563	429	239	91	24	(10)
(11)		CDD18.3	1268	1	3	13	36	107	249	314	304	182	52	7	1	(11)
(12)		CDH23.3	14154	9	24	112	330	943	2757	3895	3696	1841	482	61	6	(12)
(13)	CDH26.7	6872	1	4	22	78	302	1320	2136	2036	831	133	8	0	(13)	
(14)	Wind	WSAvg	3.4	3.6	3.9	4.3	4.3	3.5	3.5	2.7	2.6	2.7	3.2	3.6	3.5	(14)
(15)	Precipitation	PrecAvg	939	42	49	71	83	135	111	61	62	109	96	69	51	(15)
(16)		PrecMax	1478	170	158	259	285	364	310	177	192	279	279	192	169	(16)
(17)		PrecMin	482	0	0	3	6	4	19	0	0	5	3	3		(17)
(18)		PrecStd	210	37	37	46	50	87	68	47	51	69	71	54	42	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.1	27.0	29.0	32.3	33.9	37.5	40.5	42.5	37.4	32.4	27.7	23.9	(19)
(20)			MCWB	13.2	16.9	18.4	20.1	24.0	23.8	23.8	24.0	22.3	21.6	17.7	15.7	(20)
(21)		2%	DB	21.1	23.0	26.3	28.5	32.2	36.0	37.9	38.8	34.9	30.2	25.0	20.8	(21)
(22)			MCWB	12.9	15.1	16.4	18.5	23.6	23.8	23.8	24.0	23.0	21.3	17.0	14.7	(22)
(23)		5%	DB	17.7	21.0	23.9	27.1	30.2	34.7	36.9	37.2	32.8	27.8	22.5	17.6	(23)
(24)			MCWB	11.5	14.2	16.2	18.2	22.4	23.9	23.9	24.2	22.7	19.4	15.8	11.8	(24)
(25)		10%	DB	15.2	17.6	21.8	24.9	27.8	32.5	35.0	35.1	32.0	26.2	20.9	15.1	(25)
(26)			MCWB	8.9	12.4	14.9	17.5	21.2	24.2	24.4	24.4	22.4	18.9	15.4	10.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.5	19.0	20.1	22.5	25.5	26.8	26.9	26.4	25.8	24.0	20.6	18.7	(27)
(28)			MCD B	20.1	22.5	25.7	28.1	31.7	32.9	33.2	33.8	31.9	28.9	25.0	21.4	(28)
(29)		2%	WB	14.9	17.5	19.0	21.3	24.2	25.9	26.1	25.9	24.8	22.9	19.0	15.8	(29)
(30)			MCD B	18.4	20.8	23.9	25.9	29.6	32.7	33.1	33.6	31.0	27.4	22.8	19.0	(30)
(31)		5%	WB	11.9	15.5	17.9	20.2	23.4	25.3	25.6	25.5	24.0	21.8	17.6	13.5	(31)
(32)			MCD B	17.6	19.0	21.8	24.7	28.5	32.1	32.7	33.4	29.9	26.3	20.5	15.9	(32)
(33)		10%	WB	9.4	13.1	16.5	19.2	22.5	24.7	25.1	25.0	23.4	20.5	15.8	10.8	(33)
(34)			MCD B	14.4	18.0	20.5	23.6	26.9	31.0	32.2	32.7	28.6	24.0	19.8	15.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.0	11.6	12.0	12.1	10.5	10.6	11.2	11.5	11.8	12.5	12.4	11.0	(35)
(36)			MCDBR	16.5	15.7	14.3	13.7	12.0	12.5	13.3	13.6	13.5	13.7	14.8	15.3	(36)
(37)			MCWBR	9.9	9.5	7.9	7.3	5.7	4.5	3.8	3.9	4.7	6.5	8.7	10.0	(37)
(38)		5% WB	MCD BR	14.8	13.0	11.4	11.3	10.6	10.2	11.0	11.3	10.7	11.0	12.3	12.5	(38)
(39)			MCWBR	10.3	9.2	7.4	7.5	5.8	4.7	4.3	4.0	4.3	5.9	9.5	10.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.299	0.306	0.346	0.378	0.413	0.425	0.428	0.433	0.389	0.347	0.320	0.298	(40)	
(41)		taud	2.503	2.501	2.411	2.341	2.310	2.315	2.343	2.282	2.372	2.483	2.494	2.509	(41)	
(42)		Ebn at Noon	913	945	927	908	875	860	855	845	869	885	879	892	(42)	
(43)		Edh at Noon	83	93	111	125	130	129	125	131	114	94	83	78	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.79	3.36	4.43	5.45	5.84	6.64	6.76	6.17	5.11	3.91	2.96	2.38	(44)	
(45)		RadStd	0.29	0.51	0.29	0.43	0.61	0.66	0.38	0.36	0.49	0.55	0.38	0.28	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page